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SAP ABAP Data Dictionary :A Complete Beginner's Guide to Domains, Data Elements, and Database Tables



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Introduction

In SAP ABAP development, one of the most fundamental and frequently used components is the **Data Dictionary (DDIC)**. Whether you are creating reports, SmartForms, ALV reports, interfaces, or custom applications, DDIC plays a critical role in defining and managing data consistently across the SAP landscape.

The SAP Data Dictionary serves as a centralized repository that stores metadata definitions for database objects. It helps developers maintain data integrity, improve reusability, and ensure consistency throughout the system.

This article provides a comprehensive overview of SAP Data Dictionary concepts, including Domains, Data Elements, Database Tables, Structures, Table Types, and Technical Settings. By the end of this guide, you will have a solid understanding of DDIC and its importance in SAP ABAP development.

What is SAP Data Dictionary (DDIC)?

SAP Data Dictionary (DDIC) is a central repository where all data definitions and metadata are maintained.

It acts as a bridge between SAP applications and the underlying database by providing standardized definitions for business data.

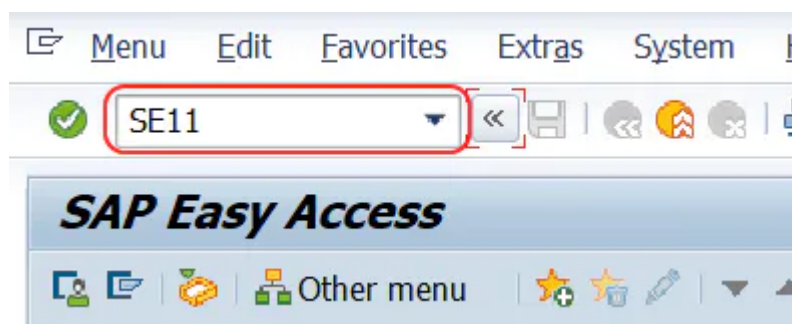
Key Benefits of DDIC

- ✓ Centralized Data Management
- ✓ Data Consistency Across Applications
- ✓ Reusability of Data Objects
- ✓ Improved Performance
- ✓ Reduced Development Effort
- ✓ Database Independence

Execute T-code or Path

From the SAP Easy Access Screen: execute transaction code SE11 or follow the SAP menu path and Execute: SE11 ABAP Dictionary

Transaction code	SE11
Menu Path	SAP Menu > Tools > ABAP Workbench > Development > SE11 – ABAP Dictionary



DDIC Architecture Overview

The relationship between DDIC objects follows a hierarchical structure:

Domain



Data Element



Table Field



Database Table

ABAP Dictionary: Initial Screen

☒ Database table

☐ View

☐ Data type

☐ Type Group

☐ Domain

☐ Search help

☐ Lock object

Domain & How to create a Domain?

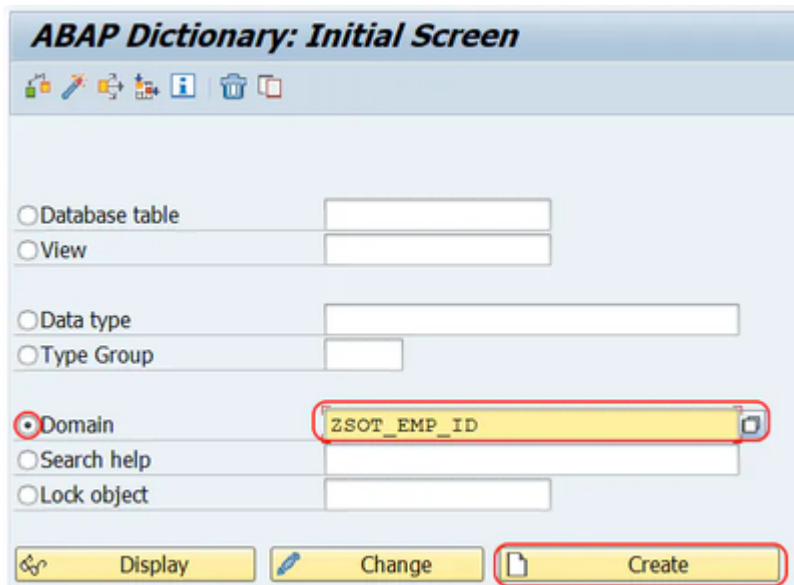
A **Domain** defines the technical attributes of a field.

It specifies:

- Data Type
- Length
- Decimal Places
- Value Range
- Fixed Values

Think of a Domain as a blueprint that determines how data should be stored in the database.

- Choose the Domain radio button
- Give the domain name that you want to create.



ABAP Dictionary: Initial Screen

Database table

View

Data type

Type Group

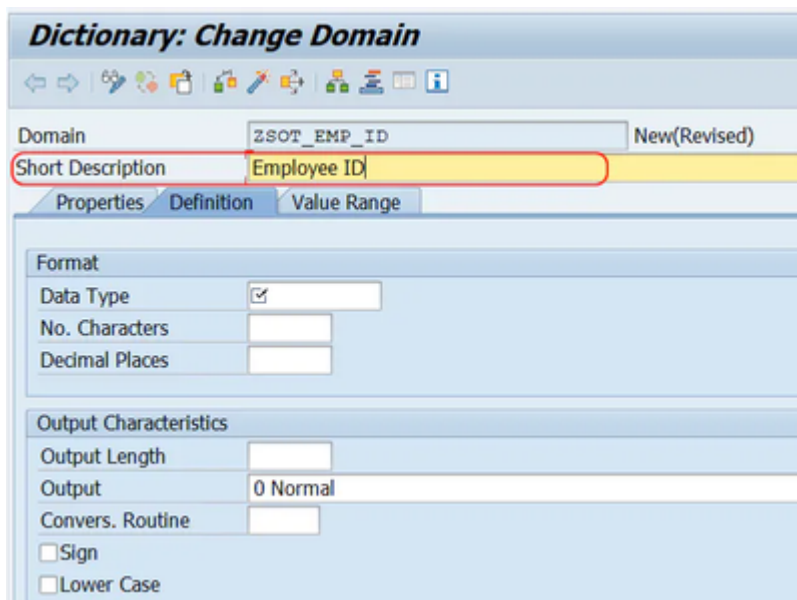
☒ Domain

Search help

Lock object

Display Change Create

Enter the short description of the domain. You can maintain text up to 60 characters.



Dictionary: Change Domain

Domain New(Revised)

Short Description

Properties Definition Value Range

Format

Data Type ☒

No. Characters

Decimal Places

Output Characteristics

Output Length

Output

Convers. Routine

☐ Sign

☐ Lower Case

On domain definition, you need to maintain the format details of the data type and length.

Dictionary: Change Domain

Domain: ZSOT_EMP_ID New(Revised)

Short Description: Employee ID

Properties Definition Value Range

Format

Data Type	NUMC	Character string with only digits
No. Characters	10	
Decimal Places	0	

Output Characteristics

Output Length	10
Convers. Routine	
☐ Sign	
☐ Lower Case	

After providing all the required details for the domain, click on the save button.

Dictionary: Change Domain

Domain: ZSOT_EMP_ID New(Revised)

Short Description: Employee ID

Properties Definition Value Range

Format

Data Type	NUMC	Character string with only digits
No. Characters	10	
Decimal Places	0	

Output Characteristics

Output Length	10
Convers. Routine	
☐ Sign	
☐ Lower Case	

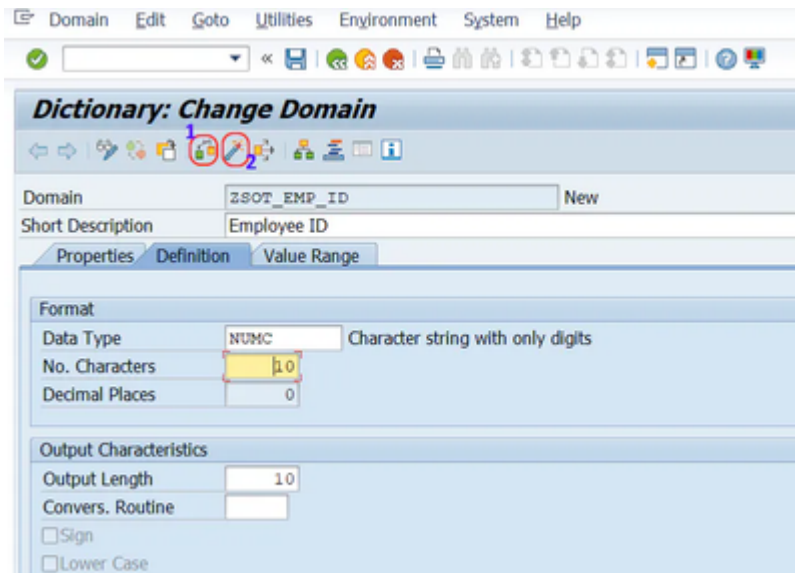
Select the local object to save the created domain in an object directory.

Dictionary: Change Domain

Transportable Objects: Local objects

D_Object	Obj_name	User
DOMA	ZSOT_EMP_ID	JAPPOER

Check Syntax & Activate.



Click **Activate (Ctrl+F3)** to activate the Domain.

Advantages of Domains

- Eliminates redundant definitions
- Improves consistency
- Easy maintenance
- Reusable across multiple applications

Data Element & How to create a Data Element?

A **Data Element** provides business meaning to a field.

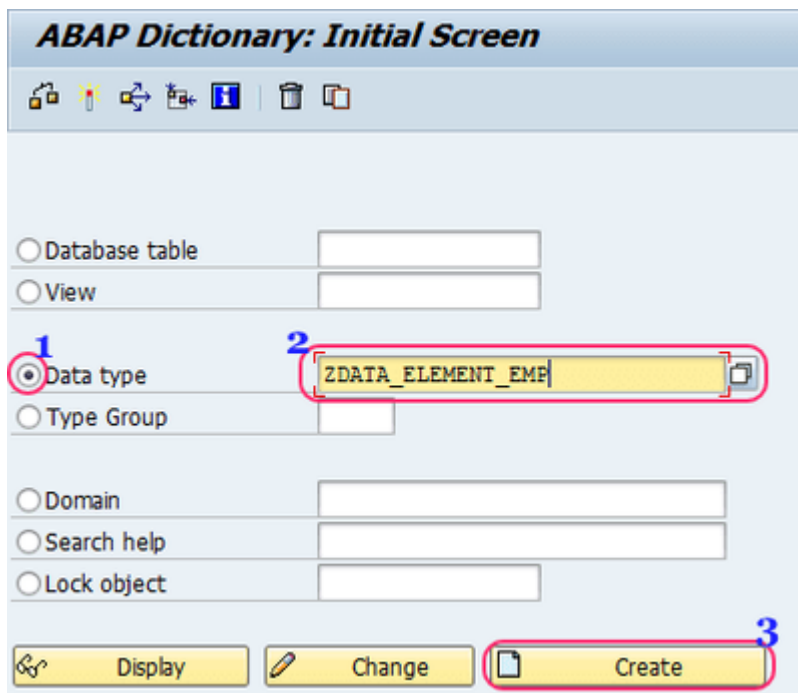
While a Domain defines technical properties, a Data Element explains what the field represents from a business perspective.

Data Element Contains

- Field Labels
- Documentation
- Business Description
- Domain Reference

The Domain controls storage and validation, while the Data Element tells users that the field represents Employee Status.

- Choose the Data Element radio button.
- Enter the Data Element name that you want to create.



ABAP Dictionary: Initial Screen

Database table

View

1 ☒ Data type **2**

Type Group

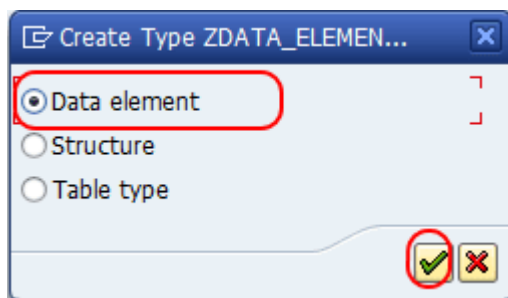
Domain

Search help

Lock object

3 Display Change Create

As we are creating data element, select data element radio button and press enter.



Create Type ZDATA_ELEMEN...

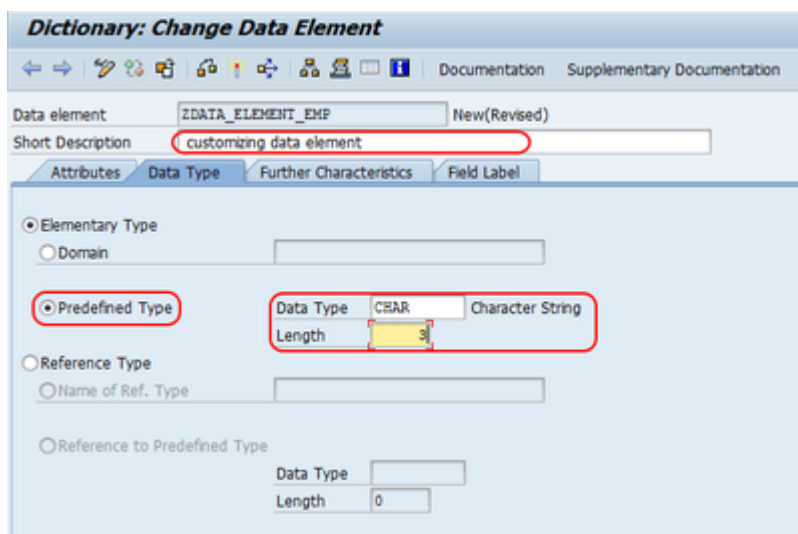
☒ Data element

☐ Structure

☐ Table type

☒ ☐

1. Update the short description of the Data Element.
2. Select the predefined radio button and update the data type, and length fields.



Dictionary: Change Data Element

Data element New(Revised)

Short Description

Attributes Data Type Further Characteristics Field Label

☒ Elementary Type

☐ Domain

☒ Predefined Type

Data Type Character String

Length

☐ Reference Type

☐ Name of Ref. Type

☐ Reference to Predefined Type

Data Type

Length

Text information is updated in field label tab, click on field label and update the following information.

Dictionary: Change Data Element

Data element: ZDATA_ELEMENT_EMP New(Revised)

Short Description: customizing data element

Attributes | Data Type | Further Characteristics | **Field Label**

	Length	Field Label
Short		
Medium		
Long	30	Employee name
Heading		

- Click on the save button (ctrl+s) to save the configured data element in a package. You get a message as “ZDATA_ELEMENT_EMP saved.
- Click **Check (Ctrl+F2)** and then **Activate (Ctrl+F3)**.

Domain vs Data Element

Domain	Data Element
Technical Definition	Business Definition
Data Type & Length	Labels & Documentation
Validation Rules	User Understanding
Reusable	Reusable

Database Tables & How to create a table?

Database Tables are used to store application data physically in the database.

Every business transaction ultimately stores information in one or more database tables.

Components of a Table

- Fields
- Primary Key
- Technical Settings
- Delivery Class
- Foreign Keys

On the initial screen, select the database table option and enter the name of the database table that you want to be created. After updating name click on create button. Here we are creating ZTEST_EMP table.

Database table ☒ View ☐ Data type Type Group Domain Search help Lock object

Display Change Create

On the next screen, update the following data. By default delivery and maintenance tab is selected.

1. **Short Description:** Update the short description of the database table.
2. **Delivery Class:** Update the delivery class from the list, you can get the list by clicking the search option (F4).
3. **Data Browser/ Table View Maintain:** Select the display/maintenance allowed from the drop down list.

Transp. Table: ZTEST_EMP New(Revised)

Short Description: Employees data table

Attributes Delivery and Maintenance Fields Entry help/check Currency/Quantity Fields

Delivery Class: A

Data Browser/Table View Maint.: Display/Maintenance Allowed with Restrictions

Select field tab and update the following data.

1. **Field:** Update the field name in the field column (16 characters)
2. **Key:** Select the key option if you want the field as table key.
3. **Data Element:** Update the data element key from the list.

Click on save button to save the configured data. You get a message as ZTEST_EMP saved.

Types of SAP Tables

1. Transparent Tables

Most commonly used table type.

Characteristics:

- One SAP table corresponds to one database table.
- Data is physically stored.

Examples:

MARA
KNA1
LFA1

2. Pooled Tables

Used to store small amounts of data.

Multiple logical tables share one physical database table.

3. Cluster Tables

Store related records together.

Primarily used in older SAP systems.

Note:

In SAP S/4HANA, Transparent Tables are predominantly used.

Key Takeaways

- Domain defines technical properties.
- Data Element provides business meaning.
- Database Tables store application data physically.
- DDIC ensures consistency and reusability across SAP applications.

Conclusion

SAP Data Dictionary (DDIC) serves as the foundation of SAP ABAP development by providing centralized definitions for business data. Understanding Domains, Data Elements, and Database Tables is essential for building scalable and maintainable SAP applications.

A strong understanding of DDIC not only helps developers create efficient solutions but also plays a significant role in SAP ABAP interviews and real-world project implementations.

****In the next article, we will explore Structures, Table Types, and Technical Settings in detail.****

LABELS

[GenAI Assisted Content](#)[Learning](#)

TAGS

[ABAP](#)[Data Dictionary](#)[Data Elements](#)[Ddic](#)[Domain](#)

Comments



venkatamandavilli Contributor

2026 Jun 17 7:24 PM

[@Pranav Sharma](#)

This is a really solid, beginner-friendly guide to the ABAP Data Dictionary (DDIC). It clearly explains the core objects and how they relate:

- **Domains** define the technical attributes: data type, length, decimals, and value range. They act as the blueprint for how data is stored.
- **Data Elements** give business meaning to a field. They use the domain for technical details but add semantics like labels and descriptions.
- **Database Tables** are built using data elements, and the hierarchy follows: Domain → Data Element → Table Field → Database Table.